CO4 - AT1 - Comparitive Analysis

Slot - B - CSA0311 - Data Structures

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1)

	Hash table	BST
Locking	Per-bucket, fine grained, low contention	Root is a hotspot, rotations need broad locks
Concurrent access	Near - linear scaling	Path to root dependency limits parallelism
Resizing	Needs rehashing	No resizing needed
Cache	Bucket array cache-friendly, chain nodes scattered	Poor locality, worsens with depth

=> Better concurrency: ~~BST~~ = Hash table, BST only wins if you need sorted/range queries.

2)

	Counting	Radix	Bucket
Non-integer keys	Not usable directly	Works via bit-reinterpret	Natural fit
Distribution sensitivity	N/A	Distribution-agnotic	Needs uniformity to shine
Memory	Not viable for floats	O(n), multi-pass	O(n), bucket overhead
Stability	Stable	stable	Stable